

SPhO 2024 Theory Paper – Worked Solutions

Q1

(a)(i)

For SHM, $v_{\max} = A\omega$. From the graph $A = 3 \text{ m}$, $v_{\max} = 2 \text{ m s}^{-1} \Rightarrow \omega = \frac{2}{3} \text{ s}^{-1}$, hence $T = 3\pi \text{ s}$.

(a)(ii)

Let $x = A \cos \theta$, $v = -A\omega \sin \theta$ with $A\omega = 2$. At A: $v = 1 \Rightarrow \sin \theta_A = -\frac{1}{2}$,
 $x > 0 \Rightarrow \theta_A = -\pi/6$. At B: $x = 0$, $v < 0 \Rightarrow \theta_B = \pi/2$. Thus $\Delta t = \frac{\theta_B - \theta_A}{\omega} = \frac{2\pi/3}{2/3} = \pi \text{ s}$.

(b)(i)

Conservation of energy:

$$\frac{1}{2}mv^2 + \frac{1}{2}m\omega^2x^2 = \frac{1}{2}m\omega^2x_0^2 \Rightarrow E_{\text{pot}}(x) = \frac{1}{2}m\omega^2x^2.$$

(b)(ii)

For horizontal displacement x , the height gain is $h = L - \sqrt{L^2 - x^2}$, so

$$E_{\text{pot}} = mgL - mg\sqrt{L^2 - x^2}.$$

(b)(iii)

For $|x| \ll L$: $\sqrt{L^2 - x^2} \simeq L - \frac{x^2}{2L}$, hence $E_{\text{pot}} \simeq \frac{mg}{2L}x^2$. Equating with $\frac{1}{2}m\omega^2x^2$ gives $\omega^2 = g/L$ and

$$T = 2\pi\sqrt{\frac{L}{g}}.$$

Q2

Given $m = 2.5 \text{ kg}$, $\omega = 3 \text{ s}^{-1}$, $r = 0.30 \text{ m}$ (then 0.50 m), $\mu_s = 0.40$.

(a)

$$f_{s,\max} = \mu_s mg = 0.4 \times 2.5 \times 9.81 = 9.81 \text{ N}.$$

(b)

$$f_{\text{req}} = m\omega^2 r = 2.5 \times 9 \times 0.3 = 6.75 \text{ N} < f_{s,\max}; \text{ no sliding.}$$

(c)

$$\mu_{\min} = \frac{\omega^2 r}{g} = \frac{9 \times 0.5}{9.81} = 0.459.$$

(d)

Slip threshold $m\omega^2 r = \mu mg$. The linear threshold speed is

$$v_{\text{crit}} = r\sqrt{\frac{\mu g}{r}} = \sqrt{\mu gr} = 1.40 \text{ m s}^{-1}.$$

Q3

(a)

$$\frac{1}{2}kd^2 = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{k}{m}}d.$$

(b)

Momentum: $MV + mv = 0 \Rightarrow V = -\frac{m}{M}v$. Energy at separation: $\frac{1}{2}kd^2 = \frac{1}{2}mv^2 + \frac{1}{2}MV^2 = \frac{1}{2}v^2\left(m + \frac{m^2}{M}\right)$. Thus

$$v = d\sqrt{\frac{kM}{m(M+m)}}.$$

(c)

Let the sphere move x and the block y (opposite directions). Spring regains length d : $x + y = d$; COM fixed: $mx = My \Rightarrow y = \frac{m}{M}x$. Hence $x = \frac{M}{M+m}d$, $y = \frac{m}{M+m}d$; the block travels $y = \frac{m}{M+m}d$.

Q4

At distance x , rod length $\ell = 2x \tan \frac{\alpha}{2}$, loop area $A = x^2 \tan \frac{\alpha}{2}$. Then $\dot{A} = 2xv \tan \frac{\alpha}{2}$, $\varepsilon = B\dot{A}$. Rod resistance $R_{\text{tot}} = R\ell$, so $I = \varepsilon/R_{\text{tot}} = Bv/R$. Magnetic drag $F = I\ell B = \frac{2B^2 \tan(\alpha/2)}{R} x v$ (opposing motion).

$$m \frac{dv}{dt} = -\frac{2B^2 \tan(\alpha/2)}{R} x v, \quad \frac{dx}{dt} = v \Rightarrow \frac{dv}{dx} = -\frac{2B^2 \tan(\alpha/2)}{mR} x.$$

Integrate from (x_0, v_0) to $(x_{\text{stop}}, 0)$ to get

$$x_{\text{stop}} = \sqrt{x_0^2 + \frac{mv_0 R}{B^2 \tan(\alpha/2)}}.$$

Q5

Let $a = \frac{R_1 + R_2}{2}$.

(a)(i)

$$v_p = \sqrt{GM \left(\frac{2}{R_1} - \frac{1}{a} \right)} = \sqrt{\frac{2GM R_2}{R_1(R_1 + R_2)}}.$$

(a)(ii)

$$\Delta E_1 = m \left(-\frac{GM}{2a} + \frac{GM}{2R_1} \right) = \frac{GMm}{2R_1} \frac{R_2 - R_1}{R_1 + R_2}.$$

(a)(iii)

$$t_{\text{Hoh}} = \pi \sqrt{\frac{a^3}{GM}} = \pi \sqrt{\frac{\left(\frac{R_1 + R_2}{2} \right)^3}{GM}}.$$

(b)

L conserved: $\omega_f = \frac{I_i}{I_f} \omega_i = \frac{110}{230} \cdot 5.2 = 2.487 \text{ s}^{-1}$.

$$K_i = \frac{1}{2} I_i \omega_i^2 = 1.487 \times 10^3 \text{ J}, \quad K_f = \frac{1}{2} I_f \omega_f^2 = 7.11 \times 10^2 \text{ J},$$

$$\Delta K \approx 7.76 \times 10^2 \text{ J}.$$

Q6

From the figure, $A = 0.80 \times 0.60 = 0.48 \text{ m}^2$, $N = 10$, $I = 5.0 \text{ A}$, $B = 0.50 \text{ T}$, $R = 0.10 \text{ m}$, $m = 5.0 \text{ kg}$.

(a)(i)

$$\mu = NIA = 24 \text{ A m}^2.$$

(a)(ii)

$$\tau_{\text{mag},0} = \mu B = 12 \text{ N m}.$$

(b)

$$\tau_g = mgR = 4.91 \text{ N m}; \tau_{\text{net}} = 12.0 - 4.91 = 7.09 \text{ N m}.$$

(c)(i)

$$\Delta U = -\mu B = -12 \text{ J}.$$

(c)(ii)

$$\Delta y = R(\pi/2) = 0.157 \text{ m}, \text{ so } mg\Delta y = 7.70 \text{ J}. K = \mu B - mg\Delta y = 4.30 \text{ J} \Rightarrow v = \sqrt{2K/m} = 1.31 \text{ m/s}.$$

Q7

(a)

$$\text{With } V = Ae^{-\alpha t} \cos \omega t, \alpha = \frac{R}{2L}, \omega = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}.$$

(b)(i)

$$\alpha = 1.70 \times 10^5 \text{ s}^{-1}, \omega \approx 5.23 \times 10^5 \text{ s}^{-1}, f = \omega/(2\pi) = 8.34 \times 10^4 \text{ Hz}, t_{0.1} = \ln 10/\alpha = 1.35 \times 10^{-5} \text{ s}.$$

(b)(ii)

$$R_c = 2\sqrt{L/C} = 2.42 \times 10^2 \Omega.$$

Q8

$$v = 0.8c, \gamma = 1/0.6 = 1.667, L = 10 \text{ ly.}$$

(a)

$$t_E = \frac{L}{v} = 12.5 \text{ yr.}$$

(b)

$$t_{\text{ship}} = t_E/\gamma = 7.5 \text{ yr.}$$

(c)

$$\text{Earth: } t = L/c = 10 \text{ yr. Ship: } t' = \gamma(t - vx/c^2) = 1.667(10 - 8) = 3.\bar{3} \text{ yr.}$$

(d)

Spacecraft at B at 12.5 yr; return light needs 10 yr to reach A $\Rightarrow$ arrival at 22.5 yr: not simultaneous.

Q9

$$P_A = 101.3 \text{ kPa, } T_A = 278.15 \text{ K, } n = 1 \text{ mol. } V_A = \frac{RT_A}{P_A} = 2.284 \times 10^{-2} \text{ m}^3.$$

(a)

$$W_{AB} = 0, W_{BC} = P_B(V_C - V_B) = \frac{1}{2}P_A V_A \approx 1.16 \times 10^3 \text{ J.}$$

(b)

$$W = nRT \ln(V_C/V_A) = RT_A \ln 2 \approx 1.60 \times 10^3 \text{ J.}$$

Q10

(a)

Total path difference for oblique incidence:

$$\Delta = d(\sin \beta + \sin \alpha).$$

Maxima when $d(\sin \beta + \sin \alpha) = m\lambda$.

(b)

For small β , $\sin \beta \approx \beta$ (rad). Then

$$\beta_m \approx \frac{m\lambda}{d} - \sin \alpha \quad \Rightarrow \quad \Delta\beta = \beta_{m+1} - \beta_m \approx \frac{\lambda}{d} \text{ (independent of } \alpha).$$